Palm Oil Retention & Melting System

Submerged Hydronic Coil Design — Architectural Revision II

Based on the *VCH Sizing Framework, Second Edition*
A Unified Design Methodology for Submerged Hydronic Coil Systems

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This document presents the second architectural revision of the palm oil storage control system, covering trunk and branch sizing, the need-array priority model, P2 governance, load-responsive operating modes, water-side hydraulics, process duty cycles, aggregate demand budgeting, and a fully specified dynamic water-temperature control law. The work was independently researched, modelled, and authored by Sambridda Ranabhat.

Contents

Executive Summary **3**

1 Trunk and Branch Sizing **4**
1.1 Rationale . . . 4
1.2 Trunk and Branch Cross-Sections . . . 4
1.3 Flow Rate at a Single Branch . . . 4
1.4 Trunk Capacity: Powering Two Branches . . . 4
1.5 Single-Branch (Reduced Trunk Velocity) Case . . . 5

2 Priority Architecture II: The Need Array **5**
2.1 From Ladder to Array . . . 5

3 PO Tank (P2) Governance: Suspicion Meter, Not Alarm **6**
3.1 Rationale for the Change . . . 6
3.2 The Suspicion Meter . . . 6
3.3 The Hard Ceiling: 2× Requested Volume . . . 6

4 Load-Responsive Operation: Eco, Efficiency, and Full Throttle Modes **7**
4.1 Mode Governed by T_5 Level . . . 7
4.2 Mode Selection vs. Dynamic Water Temperature . . . 7

5 Water-Side Hydraulics (Retained Basis) **8**

6 P1 — T_5 Duty Cycle: Worst-Case Analysis **8**
6.1 Demand Basis: 4 t/day . . . 8
6.2 Sensible Heating Demand . . . 8
6.3 Worst-Case Standing Loss . . . 8
6.4 Duty Period . . . 9

7 P2 — PO Tank Duty Cycle: Unchanged from Revision I **9**

8 P3 — Dual-Delivery Melting Window **9**
8.1 The Doubled Effective Budget . . . 9
8.2 Melt Rate (Unchanged Coil Basis) . . . 9
8.3 Mass Melted . . . 9

9 Aggregate Daily Demand and System Output **10**
9.1 Total Daily Thermal Energy Demand . . . 10
9.2 System Throughput and Fulfilment . . . 10

10 Dynamic Water Temperature Control **11**
10.1 Motivation . . . 11
10.2 Aggregate Demand Function . . . 11
10.3 From Demand to Water Temperature . . . 12
10.4 Solar and Auxiliary Gating . . . 12
10.5 Exchanger Outlet Water Temperature . . . 13

11 PLC Implementation: Processes and Logic to Insert **14**
11.1 Sensor and Signal Inputs (Tags) . . . 14
11.2 Calculated Registers . . . 14
11.3 Need-Array Arbiter . . . 14
11.4 Mode State Machine . . . 15
11.5 P1 — T_5 Block . . . 15
11.6 P2 — PO Tank Block . . . 15
11.7 P3 — Scoring and Mode Block . . . 15
11.8 Solar and Auxiliary Gating Block . . . 15
11.9 Fault Detection Block . . . 16
11.10Timers . . . 16
11.11Alarms and Interlocks . . . 16
11.12HMI Elements . . . 17

12 System Estimation and Conclusions **18**

Executive Summary

This document presents the second architectural revision of the oil storage control system, superseding the sequential, single-branch design of Revision I. The central change is a **trunk upsize** — from a 1-inch trunk feeding 1-inch branches, to a 1.5 inch trunk capable of supplying **two** 1-inch branches simultaneously at their full rated flow. This single hardware change cascades through the entire control architecture: priority is reframed as a **two-slot need array** rather than a strict ladder, the P3 (palm oil melting) process gains access to a doubled effective daily energy budget by running two branches in parallel, the PO tank's governance moves from a hard duty-cycle cap to an operator-trust model backed by a suspicion meter, and — most significantly — the water supply temperature itself becomes a **dynamic, demand-driven quantity** rather than a fixed 70 °C setpoint.

Headline Results

Metric	Result
Trunk flow advantage over a single branch	2×
T_5 (P1) daily duty period, worst case	3.67 hr (15.3%)
PO tank (P2) daily duty period	0.59 hr (2.4%)
P3 effective dual-branch energy budget	≈43.7 branch-hr/day
Palm oil melted per day (P3, dual-delivery)	≈4.92 t/day
Combined system throughput (P3 melt + PO tank)	≈5.42 t/day
Fulfilment vs. original 2 t/day mandate	≈246% (melt only)
Total daily thermal energy demand	≈1081.5 MJ
Average continuous power demand	≈12.5 kW

Key Findings

- **Dual-branch operation roughly doubles P3's effective daily throughput** relative to Revision I, without any change to coil geometry or water temperature basis — the gain comes entirely from the trunk's ability to serve two branches at once.
- **Priority is no longer a fixed ladder.** It is now expressed as a *need array*: however many of P1/P2/P3 currently have live demand determines whether the trunk runs a single branch or two branches in parallel. P2 remains capped at a fixed daily allocation; P1 and P3 are unbounded within the array.
- **P2 governance has shifted from hard cutoff to operator trust.** There is no automatic alarm-and-shutoff for exceeding expected PO-tank throughput; instead, an escalating suspicion meter and a hard 2× requested-volume ceiling govern it, with the final stop always requiring a manual action.
- **The system now runs load-responsive, not schedule-driven.** Eco, Efficiency, and Full Throttle modes adjust P3's operating posture directly off the T_5 level, and the water supply temperature is derived from real-time aggregate demand rather than held at a fixed ceiling.

- **The dynamic water-temperature control law is now fully specified.** A two-point calibration — idle at 40 °C, T_5 worst-case draw at 70 °C — resolves the low-demand inconsistency identified in the working notes and gives the target a clear physical meaning at every operating point.
- **This revision's worst-case T_5 loss calculation deliberately evaluates the tank at its 55 °C dynamic upper thermal boundary** — a conservative, worst-case basis, not an error.

On terminology. An earlier working sketch of this revision used the labels "HSM" (High Supply Mode) and "LSM" (Low Supply Mode) to describe single- vs. dual-branch operation, together with a set of lettered sub-modes. These were exploratory notation only and are **not** carried into the final architecture described here. The underlying concept — the need array determining single- vs. dual-branch operation — is retained and formalised in what follows.

1 Trunk and Branch Sizing

1.1 Rationale

Revision I's coil sizing (2-inch bore throughout, 30 m/10 m lengths) left substantial thermal margin relative to P1's and P2's actual demand — the water-side film resistance was shown to be a small fraction of overall U. This revision asks a different question of the same coils: rather than adding more thermal margin, can the *trunk* be upsized so that two branches can be served at once, letting that existing coil margin translate directly into higher system throughput?

1.2 Trunk and Branch Cross-Sections

Table 1: Trunk vs. branch sizing.

	Trunk (1.5-inch)	Branch (1-inch)
Flow area	$1.433 \times 10^{-3}\,\mathrm{m}^2$	$6.069 \times 10^{-4}\,\mathrm{m}^2$
Design velocity (branch, unchanged)	—	$1.2\,\mathrm{m\,s}^{-1}$

1.3 Flow Rate at a Single Branch

$$F_{\text{1-inch}} = A \times V = 6.069 \times 10^{-4} \times 1.2 = 7.283 \times 10^{-4}\,\mathrm{m}^3\,\mathrm{s}^{-1} \tag{1}$$

(Consistent with the $7.315 \times 10^{-4}\,\mathrm{m}^3\,\mathrm{s}^{-1}$ figure used in the working notes; small variance from rounding of the branch area.)

1.4 Trunk Capacity: Powering Two Branches

For the 1.5-inch trunk to supply **two** branches simultaneously, each at the branch design flow rate:

$$F_{\text{trunk,required}} = 2 \times F_{\text{1-inch}} \approx 1.463 \times 10^{-3}\,\mathrm{m}^3\,\mathrm{s}^{-1} \tag{2}$$

Solving for the corresponding trunk velocity:

$$V_{\text{trunk}} = \frac{F_{\text{trunk,required}}}{A_{\text{trunk}}} = \frac{1.463 \times 10^{-3}}{1.433 \times 10^{-3}} \approx 1.021\,\text{m}\,\text{s}^{-1} \tag{3}$$

The 1.5-inch trunk, run at $\approx 1.021\,\text{m}\,\text{s}^{-1}$, delivers exactly $2\times$ the flow of a single 1-inch branch at its normal design velocity. This is the hardware basis for dual-branch (parallel) operation: the trunk does not need to be oversized further than this to support two simultaneously active branches without starving either one below its individual design flow.

1.5 Single-Branch (Reduced Trunk Velocity) Case

When only one branch is active, the trunk need only match that branch's demand:

$$V_{\text{trunk,single}} = \frac{F_{\text{1-inch}}}{A_{\text{trunk}}} = \frac{7.283 \times 10^{-4}}{1.433 \times 10^{-3}} \approx 0.508\,\text{m}\,\text{s}^{-1} \tag{4}$$

(Consistent with the $\approx 0.511\,\text{m}\,\text{s}^{-1}$ figure in the working notes.) This confirms the trunk is comfortably able to throttle down to serve a single branch without any flow-control conflict.

2 Priority Architecture II: The Need Array

2.1 From Ladder to Array

Revision I resolved P1/P2/P3 contention through a strict sequential energy budget: P1 and P2 were served first, in full, and P3 received whatever remained of the day. That model assumed only one branch could ever be active at a time. With the trunk now able to serve two branches at once, priority is reframed as a **need array** — the live set of processes currently requesting water:

Table 2: Process-to-tank mapping.

Slot	Process
P1	T_5 trim-heating
P2	PO tank (C^*) maintenance / dispense
P3	T_{10} palm oil melting (remaining tanks)
P4	Reserved / unused

- **If the need array contains one active priority**, the trunk runs at single-branch (reduced) velocity, serving that process alone — behaviourally identical to Revision I's single-branch case.
- **If the need array contains two active priorities**, the trunk runs at its full dual-branch velocity (Section 1), serving both simultaneously at their individual full design flow.

- **No hard priority lockout is required between whichever two processes are concurrently active**, since the trunk can serve both at full rate — this is the key structural difference from Revision I, where a second process had to wait even though the physical hardware was not the actual bottleneck.
- P1 remains unbounded in its own demand and is effectively guaranteed service at all times; P2 remains capped (Section 3); P3 has no cap of its own beyond whatever branch-time the array leaves it, but benefits directly from dual-branch operation whenever a second slot is free.

3 PO Tank (P2) Governance: Suspicion Meter, Not Alarm

3.1 Rationale for the Change

Revision I capped P2 with a flat arbitration-window limit and treated overrun purely as a scheduling problem. This revision replaces that with an **operator-trust model**: since P2 dispensing is always operator-initiated and operator-confirmed, the system's role shifts from *preventing* overrun to *flagging* it, escalating only if the operator does not respond.

3.2 The Suspicion Meter

- Expected daily PO-tank throughput remains 500 kg/day (unchanged from Revision I). Sustained draw beyond this threshold raises a **suspicion score**, $S \in \{0, 1, 2, 3\}$, which **resets every 24-hour interval**.
- S=1**:** the HMI queries the operator directly (a confirmation prompt, not yet an alarm).
- S=2**:** the system raises an audible/visual alarm regarding the suspicion condition.
- S=3**:** the feed to the PO tank is disabled and an alarm is sounded — the only point at which this pathway forces a shutoff.
- **If there is no HMI response within 5 minutes** of a prompt, the suspicion score is automatically incremented by 1, ensuring the escalation cannot stall indefinitely on operator inaction.

3.3 The Hard Ceiling: 2× Requested Volume

Independent of the suspicion meter, a second, harder ceiling exists: for any operator-entered dispense quantity n_d (via HMI), the system permits continued draw up to $2 \times n_d$ before issuing a warning.

This ceiling is deliberately manual-shutoff only. Crossing $2 \times n_d$ produces a warning, not an automatic cutoff — consistent with the design principle that a live, operator-initiated dispense should never be silently interrupted by the PLC. The suspicion meter is the only pathway in P2's governance that can autonomously disable the feed, and even that requires either a sustained pattern of excess draw or a prolonged absence of operator response, not a single dispense running long.

4 Load-Responsive Operation: Eco, Efficiency, and Full Throttle Modes

4.1 Mode Governed by T_5 Level

P3's operating posture is tied to the live **level** of T_5 (α_S) — the process master (P1) — rather than to T_{10} tank level or any temperature reading. T_5's level is the single clearest signal of how urgently the system as a whole needs melted oil, since every downstream process line draws from T_5 directly.

Table 3: T_5 level-driven operating modes.

α_S (T_5 level)	Mode	Behaviour
$\alpha_S \geq 90\%$	**Eco Mode**	Buffer nearly full; very low need. System runs at reduced rate.
$90\% > \alpha_S \geq 50\%$	**Efficiency Mode**	Moderate, steady-state need. Standard operating rate.
$\alpha_S < 50\%$	**Full Throttle**	High need; melt and replenish as aggressively as the need array allows.

The 50% Full Throttle threshold is deliberately the same value already used elsewhere in the control architecture — it is the exact condition ($\alpha_S < 50\%$) that triggers the original batch-release logic ($T_{10} \rightarrow T_5$ HPP transfer). The moment the system independently decides to release a batch is the same moment P3's operating posture escalates to maximum urgency, so the two mechanisms reinforce each other. The 90% upper threshold is a proposed value and can be tuned during commissioning if a different Eco/Efficiency boundary proves more suitable.

This mode logic governs P3's operating posture specifically. P1 and P2 continue to run according to their own demand logic (Sections 6 and 3) regardless of T_5 level.

4.2 Mode Selection vs. Dynamic Water Temperature

Mode selection and the demand-driven water temperature of Section 10 answer two different questions and sit in a cause-and-effect relationship, not a comparison:

- **Mode selection is upstream.** It decides *how aggressively P3 should be pulling* — a decision about *production posture*.
- $E_{\text{total}} \rightarrow T_{\text{water}}$ **is downstream.** Whatever P1, P2, and P3 actually demand — P3's demand shaped by whichever mode it is in — gets summed into E_{total}, which then decides how hot the water must be to satisfy that combined load. This is a decision about *delivery temperature*.

Concretely: Full Throttle mode increases P3's draw, which increases E_α, which raises E_{total}, which raises T_{water}. Mode *feeds into* E_{total}; it does not compete against it. There is no "select whichever is higher" step anywhere in this architecture.

5 Water-Side Hydraulics (Retained Basis)

The branch-side hydraulics, coil geometry, and overall U-values are carried forward unchanged from Revision I — the trunk upsize affects how many branches can run concurrently, not the thermal performance of any individual branch.

Table 4: Retained overall U-values (2-inch bore, unchanged coil lengths).

Duty	Basis	U (W/m^2·K)
T_5 trim-heat (liquid, $h_o = 138.9\,\mathrm{W/m^2K}$)	10 m coil	112.6
PO tank maintenance (liquid, $h_o = 138.9\,\mathrm{W/m^2K}$)	30 m coil	112.5
T_{10} palm oil melt (solid, $h_o = 35\,\mathrm{W/m^2K}$)	30 m coil	33.0

$$C_w = 2740.7\,\mathrm{W\,K^{-1}} \quad \text{(per active branch, at } V = 1.2\,\mathrm{m\,s^{-1}}\text{, unchanged from Revision I)} \tag{5}$$

$$\varepsilon_{T5}\,C_w = 205.3\,\mathrm{W\,K^{-1}} \qquad \varepsilon_{PO}\,C_w = 570.5\,\mathrm{W\,K^{-1}} \qquad \varepsilon_{T10}\,C_w = 181.5\,\mathrm{W\,K^{-1}} \tag{6}$$

6 P1 — T_5 Duty Cycle: Worst-Case Analysis

6.1 Demand Basis: 4 t/day

This revision evaluates T_5 against a **doubled** daily throughput of **4 t/day** (up from 2 t/day in Revision I), reflecting the higher system-level throughput the dual-branch architecture is now sized to support.

$$Q_{T5}(40\,^\circ\mathrm{C}) = 205.3 \times (70 - 40) = 6160\,\mathrm{W} \quad \text{(max draw)} \tag{7}$$

6.2 Sensible Heating Demand

$$E_{\text{sensible}} = 4000 \times 2100 \times 5 = 42.0\,\mathrm{MJ} \tag{8}$$

6.3 Worst-Case Standing Loss

This revision deliberately evaluates T_5's standing loss at its $55\,^\circ\mathrm{C}$ dynamic upper thermal boundary (the safety cut-off defined in the original control architecture, $T_f(x, y, m_p) > 55\,^\circ\mathrm{C}$), rather than at the $42.5\,^\circ\mathrm{C}$ operating mid-band used in Revision I. This is a conservative worst-case basis: it asks whether the system still holds up if T_5 is sitting at its hottest permissible boundary. $\Delta T = 55 - 18.1 \approx 37\,\mathrm{K}$.

$$E_{\text{loss}} = 0.686 \times 14.25 \times 37 \times 86{,}400 \approx 32.4\,\mathrm{MJ} \tag{9}$$

$$\boxed{E_{P1} = 42.0 + 32.4 = 74.4\,\text{MJ/day}} \tag{10}$$

6.4 Duty Period

The delivery rate is evaluated at the operating mid-band (42.5 °C), consistent with Revision I — only the loss term uses the worst-case boundary temperature:

$$\boxed{t_{P1} = \frac{74.4 \times 10^6}{205.3 \times 27.5} \approx 13{,}199\text{ s} \approx \mathbf{3.67}\text{ hr } (\mathbf{15.3}\%\text{ of }\mathbf{24}\text{ hr})} \tag{11}$$

7 P2 — PO Tank Duty Cycle: Unchanged from Revision I

The PO tank's own thermal demand is unaffected by the trunk upsize or the governance changes of Section 3 — those changes govern *how* overrun is handled, not the underlying 500 kg/day, 24–30 °C energy balance itself.

$$E_{P2} = 51.7\,\text{MJ/day} \qquad \boxed{t_{P2} \approx 2107\text{ s} \approx \mathbf{0.585}\text{ hr } (\mathbf{2.4}\%\text{ of }\mathbf{24}\text{ hr})} \tag{12}$$

8 P3 — Dual-Delivery Melting Window

8.1 The Doubled Effective Budget

Modelling approach: because the trunk can serve two branches at once, P3 is not limited to whatever real-time hours remain after P1 and P2 finish — it can run *two* T_{10} coils in parallel during that remaining time. This is modelled as an effective **48 branch-hour** daily pool (2 branches × 24 hr), from which P1's and P2's own single-branch consumption is subtracted. The remainder is P3's effective budget, realisable either as two tanks melting in parallel for half the real time, or one tank melting for the full equivalent duration — the total energy delivered is the same either way.

$$t_{P3,\text{budget}} = 48 - (t_{P1} + t_{P2}) = 48 - (3.67 + 0.585) \approx \mathbf{43.7}\text{ branch-hr/day} \tag{13}$$

8.2 Melt Rate (Unchanged Coil Basis)

$$Q_{T10}(36\,^\circ\text{C}) = 181.5 \times (70 - 36) \approx 6172\,\text{W} \tag{14}$$

Same $NTU_{T10} = 0.0685$, $\varepsilon_{T10} = 0.0662$ as Revision I — coil geometry and water temperature basis for this duty are unchanged.

8.3 Mass Melted

$$E_{P3} = 6172\text{ W} \times (43 \times 3600)\text{ s} \approx 955.4\,\text{MJ} \tag{15}$$

$$\boxed{m_{P3} = \frac{955.4 \times 10^6}{194{,}200} \approx \mathbf{4919.8}\ \text{kg/day}\ (\mathbf{4.92}\ \text{t/day})} \tag{16}$$

Cross-check: melting two full 10 kL tanks in parallel ($m = 17\,800$ kg) at twice the coil power (2×6172 W) takes the *same* 77.8 hr as melting one tank alone at single-coil power — mass and power both double, so melt time is unchanged. This confirms the dual-branch gain is a genuine doubling of throughput, not merely a faster single-tank cycle.

9 Aggregate Daily Demand and System Output

9.1 Total Daily Thermal Energy Demand

$$E_{\text{total}} = E_{P1} + E_{P2} + E_{P3} = 74.4 + 51.7 + 955.4 \approx 1081.5\,\text{MJ/day} \tag{17}$$

$$P_{\text{avg}} = \frac{1081.5 \times 10^6}{86{,}400} \approx 12.5\,\text{kW continuous average} \tag{18}$$

9.2 System Throughput and Fulfilment

Table 5: Combined daily throughput.

Stream	Daily mass
Palm oil melted (P3, dual-delivery)	4919.8 kg
Palm olein dispensed (P2)	500.0 kg
Combined system throughput	**5419.8 kg (≈5.42 t/day)**

$$\text{Fulfilment (melt only)} = \frac{4919.8}{2000} \times 100\% \approx \mathbf{246}\% \tag{19}$$

$$\text{Fulfilment (combined)} = \frac{5419.8}{2000} \times 100\% \approx \mathbf{271}\% \tag{20}$$

Against the original 2 t/day **mandate, this revision delivers roughly** 2.5× **to** 2.7× **the required throughput**, a direct consequence of the trunk's dual-branch capability rather than any change to individual coil sizing.

Note on the ≈6 t/day estimate. The rigorous figure worked through here is ≈5.42 t/day, slightly below the ≈6 t/day ballpark from early scoping. This gap is not an error — it reflects a **deliberately conservative modelling choice**: P3's 48 branch-hour pool is reduced by the full raw duration of P1's and P2's duty periods ($t_{P1} + t_{P2} = 4.25$ branch-hr), regardless of how much overlap actually occurs day to day. A model that credits P3 for partial overlap (P3 usually still has one branch free while P1 or P2 draws its other slot) would recover most of the remaining gap. The figure adopted here is the conservative floor by design, not the best-case ceiling.

10 Dynamic Water Temperature Control

10.1 Motivation

Both prior revisions held the water supply at a fixed 70 °C. This is now identified as unnecessary: the water only needs to be as hot as the *current* aggregate demand requires. This section formalises a control law that derives the water target from real-time load rather than a fixed setpoint, then gates the solar loop and auxiliary driver against that dynamic target.

10.2 Aggregate Demand Function

The total instantaneous energy demand is the sum of whichever processes are currently active:

$$E_{\text{total}} = E_\alpha + E_5 + E_{C^*} \tag{21}$$

E_α — Generic T_{10} Melting-Tank Demand

For a tank at current temperature t_α and mass m_α, using $C_{p,s} = 2000$, $L_f = 150{,}000$, $C_{p,l} = 2100$ J/kg.K, melt point 36 °C, delivery target 40 °C:

$$E_\alpha(t_\alpha, m_\alpha) = \begin{cases} \left[2000(36 - t_\alpha) + 150{,}000 + 2100 \times 4\right] m_\alpha, & t_\alpha \le 36\,^\circ\text{C} \\ 2100\,(40 - t_\alpha)\, m_\alpha, & 36\,^\circ\text{C} < t_\alpha \le 40\,^\circ\text{C} \\ 0, & t_\alpha > 40\,^\circ\text{C} \end{cases} \tag{22}$$

The step at exactly 36 °C is a genuine physical discontinuity, not a modelling error. Just below the melt point, the tank still requires the full 150 000 J/kg of latent heat (remaining demand ≈ 158 400 J/kg); once that heat has been absorbed, remaining demand drops sharply to liquid-phase sensible heating only (≈ 8400 J/kg). Standing heat loss is deliberately **excluded** from E_α — it is already accounted for as a separate daily energy term in Sections 8 and 9, and including it here would double-count it.

E_5 — T_5 Trim-Heating Demand

$$E_5(T_F, T_5, M_5, T) = 2100\, M_5\, (T_F - T_5) + 0.686 \times 14.25\, (T_5 - T) \tag{23}$$

where T_F is the incoming oil feed temperature and T is ambient. Unlike E_α, E_5 carries its own standing-loss term explicitly ($0.686 \times 14.25 = 9.78\,\mathrm{W\,K^{-1}}$, $\mathrm{T_5}$'s insulated jacket coefficient×area from Section 6), since $\mathrm{T_5}$'s loss was not separately budgeted elsewhere in this instantaneous-demand context the way $\mathrm{T_{10}}$'s was.

E_{C^*} — PO Tank Demand

$$E_{C^*}(T_c, M_c, T) = 2100\, M_c\,(30 - T_c) + 59.08\,(T_c - T) \tag{24}$$

The loss coefficient $59.08\,\mathrm{W\,K^{-1}}$ is $U_{\mathrm{cyl}}A_{\mathrm{cyl}} + U_{\mathrm{base}}A_{\mathrm{base}}$ from the PO tank loss calculation. This term is essential: the PO tank's standing loss outweighs its sensible duty by roughly 7:1 (tank is uninsulated), and omitting it would badly understate instantaneous demand.

On the linearity assumption in the loss coefficients. Both $9.78\,\mathrm{W\,K^{-1}}$ ($\mathrm{T_5}$) and $59.08\,\mathrm{W\,K^{-1}}$ (PO tank) were derived from natural-convection correlations evaluated at one specific ΔT. Physically, h scales roughly as $\Delta T^{1/4}$ in the laminar regime, so writing loss as a simple linear function of $(T_{\mathrm{tank}} - T)$ is an approximation, not an exact relation. It was kept deliberately: (1) it allows real-time evaluation in a 10 s-cycle PLC without re-solving a Nusselt correlation each cycle, and (2) the error is small over the narrow operating bands each tank spans — $\mathrm{T_5}$: 40–45 °C; PO tank: 24–30 °C. Insulating the PO tank (recommended in Section 3) would substantially reduce the magnitude of the loss term and make this approximation safer regardless.

Assembled Total

$$\boxed{E_{\mathrm{total}}(t_\alpha, m_\alpha, T_F, T_5, M_5, T_c, M_c, T) = E_\alpha + E_5 + E_{C^*}} \tag{25}$$

10.3 From Demand to Water Temperature

Demand E_{total} is evaluated on a 10 s refresh window and converted to an instantaneous power P. The water target temperature is then proportional to P, anchored at **two known endpoints** — idle and $\mathrm{T_5}$'s worst-case draw:

$$T_{\mathrm{water}} = 40 + \frac{70 - 40}{6.16}\,P = 40 + 4.87\,P\ [\mathrm{kW}], \quad \text{clamped to } [40\,^\circ\mathrm{C},\ 70\,^\circ\mathrm{C}] \tag{26}$$

where 6.16 kW is $\mathrm{T_5}$'s own established worst-case draw ($Q_{T5}(40\,^\circ\mathrm{C})$, Section 6). Anchoring at both endpoints — idle ($P = 0 \to 40\,^\circ\mathrm{C}$) and $\mathrm{T_5}$ worst case ($P = 6.16\,\mathrm{kW} \to 70\,^\circ\mathrm{C}$) — resolves the low-demand inconsistency of an earlier proportional law through the origin.

At $P = 1\,\mathrm{kW}$: $T_{\mathrm{water}} = 44.9\,^\circ\mathrm{C}$ — a sensible result just above the floor. At $P \geq 6.16\,\mathrm{kW}$: the target saturates at 70 °C, meaning the system reaches full design temperature any time aggregate demand is comparable to a single $\mathrm{T_5}$ worst-case event.

10.4 Solar and Auxiliary Gating

With the dynamic target T_w established, the solar loop and auxiliary driver are gated purely on whether each can meet it:

- $T_{\mathrm{solar}} > T_w$: run the through-solar cycle (solar alone meets current demand).

- $T_{\text{solar}} < T_w$: bypass solar.
- The auxiliary driver activates **only** when the system's computed demand requires it — not on a fixed schedule. Whenever the auxiliary is active, $T_w > T_{\text{solar}}$ holds by definition, so solar bypass is always concurrent with auxiliary operation.

10.5 Exchanger Outlet Water Temperature

The water returning from each exchanger retains most of its heat, since the low per-pass effectiveness of the batch-heating model means only a small fraction of the available driving temperature is extracted per loop:

$$T_{\text{water,out}} = T_{\text{hot}} - \varepsilon\,(T_{\text{hot}} - T_{\text{tank}}) \tag{27}$$

evaluated at $T_{\text{hot}} = 70\,^{\circ}\text{C}$, using $\varepsilon_{T5} = 0.0749$ and $\varepsilon_{T10} = 0.0662$.

Table 6: T_5 outlet water temperature — worst and best case.

Condition	T_{tank}	$T_{\text{water,out}}$
Worst (just triggered, P1 fires)	40 °C	67.75 °C
Best (near target, P1 about to clear)	44 °C	68.05 °C

Table 7: T_{10} outlet water temperature — worst, ambient, and best case fill.

Condition	T_{tank}	$T_{\text{water,out}}$
Worst (extreme cold winter fill)	2 °C	65.50 °C
Ambient (Kathmandu annual average)	18 °C	66.56 °C
Best (mild fill condition)	24 °C	66.96 °C

All five outlet temperatures sit within roughly 2–4.5 K of the 70 °C supply. This is the expected consequence of the deliberately low per-pass effectiveness — only a small fraction of the available driving temperature is extracted per coil pass, by design. The return line consequently carries most of its heat back to the thermal battery rather than losing it to the tank, which is consistent with and confirms the batch-heating assumption used throughout this document.

Interpretation of the T_{10} reference points. 18 °C matches the Kathmandu weighted annual ambient average used throughout this document. 2 °C is an extreme cold-season worst-case fill, and 24 °C a mild best-case fill. These are bounding reference points for this outlet-temperature sweep only — confirm this reading before treating the 2/24 °C figures as design inputs elsewhere.

11 PLC Implementation: Processes and Logic to Insert

This section consolidates every process, register, timer, alarm, and HMI element required to implement the architecture described in this document on the PLC. It is organised the way a scan cycle would actually need it — inputs first, then derived values, then the control blocks that consume them — rather than in the order each concept was introduced earlier in the document.

11.1 Sensor and Signal Inputs (Tags)

Table 8: Required PLC input tags.

Category	Tags
Temperatures	θ_S (T_5); θ_{10A}, θ_{10B}, θ_{10C} (T_{10} tanks, including the PO tank's own sensor — still read continuously even while isolated); θ_{Sol} (solar loop); θ_H (thermal battery)
Levels	α_S (T_5); α_{10A}, α_{10B}, α_{10C}
Downstream demand	δ_X, δ_Y, δ_Z (process line HMIs)
Valve/pump feedback	Ball valve open/closed limit switches (all tanks; the PO tank's is safety-critical); automated valve position feedback; trunk pump/VFD speed feedback
HMI inputs	PO tank designation selector; dispense quantity entry n_d; "process complete" acknowledgment; fault-alarm stop/continue button (Section 11.9)

11.2 Calculated Registers

Table 9: Real-time PLC-computed values.

Register	Purpose
$T_f(x, y, m_p)$	Dynamic upper thermal boundary (T_5 safety cutoff)
S_{casual}, $S_{\text{immediate}}$	Per non-PO T_{10} tank scoring
E_α, E_5, E_{C^*}, E_{total}	Refreshed on the 10 s window (Section 10)
T_{water} target	Clamped 40–70 °C
Need-array active count	How many of P1/P2/P3 currently have live demand → trunk velocity setpoint
Mode register	Eco / Efficiency / Standard / Full Throttle, from the hysteresis ladder (Section 4)
Suspicion score $S \in \{0, 1, 2, 3\}$	Per-event counter, own 24 hr reset clock (Section 3)

11.3 Need-Array Arbiter

The arbiter evaluates which of P1/P2/P3 currently have live demand and sets the trunk to single- or dual-branch velocity accordingly (Section 2). With only two branches available, a rule is required for the case where all three processes want service at once:

Three-way contention rule: keep the top priorities, discard the lowest until a slot frees. If P1, P2, and P3 all have live demand simultaneously, P1 and P2 are served (the two highest-ranked), and P3 is held back — deferred, not denied — until one of the two active slots becomes free. This is consistent with P3's role throughout this architecture as the remainder-absorbing process: it already tolerates deferral better than a process-critical T_5 event or a live, operator- supervised P2 dispense, so it is the correct process to hold back under genuine three-way contention.

11.4 Mode State Machine

Implements the hysteresis ladder of Section 4 directly: four states (Eco, Efficiency, Standard, Full Throttle), transitions gated by the rising/falling thresholds in that section's table, each with a 5 % deadband. The mode register output feeds P3's operating rate setpoint, which in turn feeds E_α (Section 10).

11.5 P1 — T_5 Block

- $\theta_S < 40\,°C$ triggers the P1 heat-demand flag (open coil valve, request a branch from the arbiter).
- T_f interlock: continuously evaluate $T_f(x, y, m_p)$; if it exceeds 55 °C, throttle or divert flow.
- Batch-release condition: $\alpha_S < 50\,\%$ **and** the highest-ranked non-PO T_{10} tank $\geq 36\,°C$ → trigger the HPP transfer.

11.6 P2 — PO Tank Block

- On HMI entry of n_d: compute dispense duration, open the dispense path, pre-allocate heat for up to $2 \times n_d$ (Section 3).
- On reaching the computed n_d volume: enter pulse-hold — coil heating loop only, 20 s on per 60 s window — until the operator registers "process complete."
- Continuously compare actual draw against $2 \times n_d$; on crossing, issue a warning and increment S by 1 (Section 3).
- Suspicion state machine: S=1 → HMI confirmation prompt; S=2 → alarm; S=3 → disable feed **and** alarm; no HMI response to a prompt within 5 min → auto-increment S.

11.7 P3 — Scoring and Mode Block

Compute and rank $S_{\text{casual}}/S_{\text{immediate}}$ across the two non-PO T_{10} tanks (Section 4 of the base VCH framework, unchanged); select the highest-ranked tank; apply the current mode's operating rate (Section 4) to that tank's melting process.

11.8 Solar and Auxiliary Gating Block

Compare θ_{Sol} against the dynamic T_{water} target: through-solar / bypass-solar / quick-solar-cycle selection, auxiliary element firing, and the thermal battery recharge interlock — all as specified in Section 10 and the base VCH control architecture.

11.9 Fault Detection Block

Implements the physics-based alarm philosophy established in Architectural Revision I: the PLC continuously compares each tank's *observed* heating and cooling rate against the *maximum physically achievable* rate given the coil's known delivery capacity and that tank's own computed standing-loss rate.

On detecting an unnatural rate (heating or cooling faster than is physically possible given known coil output and loss rate):

1. Sound the alarm immediately.
2. Present a stop button on the HMI.
3. **If the operator presses stop**, halt the affected process.
4. **If the operator does not press stop**, the system *continues operating* despite the detected anomaly — the alarm remains active/latched as a standing notice, but no autonomous shutdown is forced.

This is deliberately advisory rather than enforcing, and is consistent with the operator-trust principle already applied to P2's governance (Section 3): the PLC's role is to surface a detected anomaly clearly and immediately, not to unilaterally decide the response on the operator's behalf. The decision to continue running under a known, unresolved fault condition is deliberately left with the person who can see the physical plant, not automated away.

11.10 Timers

- PO tank pulse-hold: 20 s on / 60 s repeating.
- Suspicion meter: 24 hr reset; 5 min no-HMI-response auto-increment.
- $E_{\text{total}}/T_{\text{water}}$ refresh: 10 s.
- Thermal recharge sequence: circulate-to-36 °C, then full-auxiliary, per the base VCH recharge interlock.

11.11 Alarms and Interlocks

- T_f boundary exceeded → T_5 interlock.
- $\theta_H < 60\,^{\circ}\text{C}$ → recharge alarm sequence.
- T_{10} high-limit $> 75\,^{\circ}\text{C}$ → valve close + alarm.
- Physics-based heating/cooling rate violation (Section 11.9) → alarm + operator stop/continue choice.
- Suspicion meter S=2 (alarm), S=3 (disable + alarm).
- PO tank ball-valve/designation mismatch: cross-check that an isolated tank's manual valve has not been reopened outside the designated dispense workflow.

11.12 HMI Elements

- PO tank designation control, dispense quantity entry (n_d), process-complete button.
- Suspicion meter status and prompts at each tier.
- Mode indicator (read-only, four-state).
- Fault alarm stop/continue button (Section 11.9).
- Manual override panel, valve status readouts.
- Trend logging for θ, α, E_{total}, T_{water} — useful for commissioning verification against the figures worked through in this document.

12 System Estimation and Conclusions

1. **The trunk upsize is the single highest-leverage change in this revision.** Without altering any coil, tank, or water temperature specification, enabling two branches to run in parallel roughly doubles P3's effective daily throughput, lifting combined system output to ≈5.42 t/day (≈271% of the original 2 t/day mandate).

2. **The need-array priority model is a genuine simplification over a strict ladder**, since it lets the hardware's actual capacity (two branches) rather than an artificial single-branch assumption govern how much of the system runs concurrently.

3. **P2's shift to an operator-trust governance model is appropriate given its use case**: a live, technician-supervised dispense should not be silently interrupted, and the suspicion meter plus $2 \times n_d$ ceiling together provide layered but non-disruptive oversight.

4. **The dynamic water-temperature control law is now fully specified.** Anchoring T_{water} at both known endpoints — idle ($P = 0 \rightarrow 40\,^{\circ}\text{C}$) and T_5's own worst-case draw ($P \approx 6.16\,\text{kW} \rightarrow 70\,^{\circ}\text{C}$) — resolves the low-demand inconsistency identified during development, and gives the water target a clear physical meaning: the supply reaches full design temperature once demand is comparable to a single T_5 worst-case event, and only backs off when the system is genuinely under-loaded.

5. **Mode selection (Eco/Efficiency/Full Throttle) and the demand-driven water temperature are not competing signals.** Mode, governed by T_5's level, decides how aggressively P3 should be pulling; that decision shapes E_α, which feeds into E_{total}, which sets T_{water}. The relationship is causal, not comparative — there is no "select the higher value" step anywhere in this architecture.

6. **The worst-case-versus-typical distinction in the T_5 loss calculation should be applied consistently going forward.** This revision evaluates T_5's loss at its 55 °C dynamic boundary specifically to stress-test the architecture; day-to-day operational figures will typically run below this, and the distinction should be labelled explicitly wherever this calculation is reused, to avoid it being mistaken for the typical operating case. The same conservative-by-design principle applies to P3's 48 branch-hour budget calculation (Section 9), which is deliberately a floor estimate rather than a best case.

7. **Remaining open items for a future revision:** (a) confirm the 2/24 °C T_{10} outlet-temperature reference points (Section 10.5) against their intended design meaning; (b) consider insulating the PO tank (Section 3's standing recommendation), which would both improve efficiency and reduce the sensitivity of E_{C^*}'s linearised loss term to operating temperature; (c) if a less conservative P3 throughput estimate is desired, revisit the overlap-based refinement considered during this revision's development (crediting P3 for the fact that P1 and P2 rarely draw at the exact same instant) as an alternative to the flat floor-subtraction currently adopted.

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